

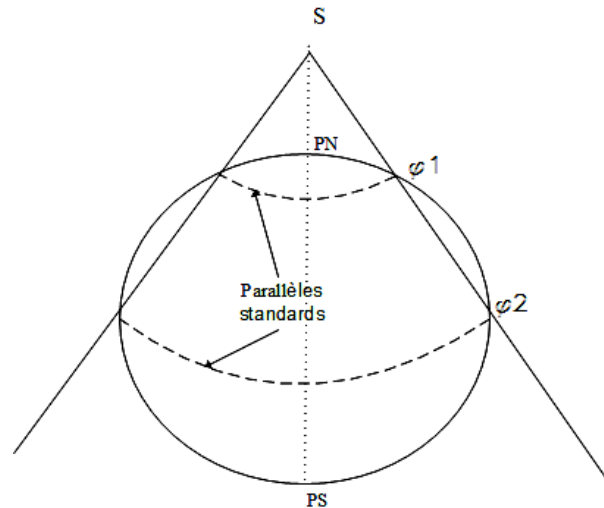
Cas d'une projection conique conforme à deux parallèles standard

Le cône est sécant en deux parallèles φ_1 et φ_2 (Figure 3.4). L'échelle est conservée sur ces deux parallèles ($K_{\varphi_1}=K_{\varphi_2}=1$).

$$\text{Donc } k_1 = k_2 = \frac{\sqrt{\left(\frac{\delta y}{\delta U_1}\right)^2 + \left(\frac{\delta x}{\delta U_1}\right)^2}}{N_1 \cos \varphi_1} = \frac{\sqrt{\left(\frac{\delta y}{\delta U_2}\right)^2 + \left(\frac{\delta x}{\delta U_2}\right)^2}}{N_2 \cos \varphi_2} = 1 \quad (3.25)$$

$$\text{On a } \begin{cases} x = \rho \sin \gamma = \rho_0 e^{-n\overline{U}_1} \sin \gamma \\ y = \rho_0 - \rho_0 e^{-n\overline{U}_1} \cos \gamma \end{cases}$$

$$\begin{cases} \frac{\partial x}{\partial U_1} = -\frac{\partial y}{\partial \lambda} = -\rho_0 n e^{-n\overline{U}_1} \cdot \sin \gamma \\ \frac{\partial x}{\partial \lambda} = \frac{\partial y}{\partial U_1} = \rho_0 n e^{-n\overline{U}_1} \cdot \cos \gamma \end{cases} \quad \text{et} \quad \begin{cases} \frac{\partial x}{\partial U_2} = -\frac{\partial y}{\partial \lambda} = -\rho_0 n e^{-n\overline{U}_2} \cdot \sin \gamma \\ \frac{\partial x}{\partial \lambda} = \frac{\partial y}{\partial U_2} = \rho_0 n e^{-n\overline{U}_2} \cdot \cos \gamma \end{cases}$$



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Figure 3.4 : Projection conique conforme avec deux parallèles standard

$$\text{L'équation (3.25) permet d'écrire } k_1 = k_2 = \frac{\sqrt{\rho_0^2 n^2 e^{-2n\overline{U}_1}}}{N_1 \cos \varphi_1} = \frac{\sqrt{\rho_0^2 n^2 e^{-2n\overline{U}_2}}}{N_2 \cos \varphi_2} = 1$$

$$\text{D'où : } \frac{e^{-n\overline{U}_1}}{e^{-n\overline{U}_2}} = \frac{N_1 \cos \varphi_1}{N_2 \cos \varphi_2}$$

$$\ln \frac{e^{-n\overline{U}_1}}{e^{-n\overline{U}_2}} = \ln(N_1 \cos \varphi_1) - \ln(N_2 \cos \varphi_2)$$

$$-n\overline{U}_1 + n\overline{U}_2 = \ln(N_1 \cos \varphi_1) - \ln(N_2 \cos \varphi_2)$$

$$n = \frac{\ln(N_1 \cos \varphi_1) - \ln(N_2 \cos \varphi_2)}{\overline{U}_2 - \overline{U}_1} \quad (3.26)$$

$$\text{Donc } \rho_0 = \frac{N_1 \cos \varphi_1}{n e^{-n\overline{U}_1}} = \frac{N_2 \cos \varphi_2}{n e^{-n\overline{U}_2}} \quad (3.27)$$